

# Benchmarking PatchTST for ICU Sepsis Onset Prediction

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## Abstract

Patients in intensive care units are continuously monitored by machines that record vital signs like heart rate, blood pressure, and oxygen levels as often as every two minutes. Combined with laboratory results and medication records, this produces an enormous amount of data for each patient over the course of their stay. Clinicians use this information to make life-or-death decisions, but the sheer volume makes it difficult to identify early warning signs of deterioration before they become emergencies.

Machine learning models have been developed to help with this problem by scanning patient data and flagging those at risk of events like circulatory failure. However, previous studies have shown that current deep learning models struggle with this type of data. Despite having access to hours or days of patient history, they perform no better, and sometimes worse than simpler methods that only look at the most recent measurements.

Our project investigates whether a newer type of model called PatchTST can do better. Instead of trying to process every single data point individually, PatchTST groups consecutive measurements into meaningful chunks. For example, one hour of heart rate readings becomes a single unit. The model then looks at relationships between these chunks rather than between individual data points. This is both more efficient and more aligned with how clinicians think about patient trajectories: trends over hours matter more than any single two-minute reading.

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## 1. Introduction

Sepsis is a leading cause of in-hospital mortality, and its case-fatality rises measurably with every hour of delayed intervention [1]. Bedside recognition is difficult because the physiological signature of onset is distributed across many weakly predictive variables evolving on different timescales, which has motivated a substantial body of work on machine-learning early-warning systems that consume the routine ICU electronic health record (EHR) stream.

This report addresses the sepsis-onset prediction task defined by the HiRID-ICU-Benchmark [2] on the HiRID dataset [3], using the Yet-Another-ICU-Benchmark framework (YAIB) [4] for cohort construction, preprocessing, and cross-validation. The task is per-timestep binary classification on an hourly grid: given a patient's multivariate ICU trajectory up to time  $t$ , predict whether Sepsis-3 criteria [5] will be met within the next six hours.

Although the HiRID-ICU-Benchmark literature has evaluated LSTM, GRU, TCN, and encoder-only Transformer baselines on this task, the patch-based time-series Transformer PatchTST [6] — which has achieved state-of-the-art performance on long-horizon forecasting benchmarks — has not, to our knowledge, been evaluated on the HiRID-ICU-Benchmark sepsis task. PatchTST's two distinguishing design choices (patching long sequences into contiguous tokens and modelling channels independently) are plausibly well-matched to autocorrelated multivariate physiological

signals. We therefore integrated PatchTST into YAIB, tuned it with a 30-trial Gaussian-Process Bayesian optimization sweep on GPU-accelerated compute, and benchmarked it against the standard HiRID-ICU-Benchmark baseline set under identical cohort, preprocessing, and  $5 \times 5$  nested cross-validation conditions.

## 2. Background and Literature Review

**Sepsis and the HiRID task.** The Sepsis-3 consensus definition operationalizes sepsis as a suspected or confirmed infection together with a two-point increase in the Sequential Organ Failure Assessment (SOFA) score. The HiRID dataset comprises roughly 33,900 de-identified ICU stays from the University Hospital of Bern, preserved at their native two-minute resolution; HiRID-ICU-Benchmark provides a canonical hourly-resampled task definition that we use directly. Prior ICU sepsis-prediction literature achieves AUROCs in the 0.80–0.85 range on MIMIC-II/III with classical methods and recurrent networks [7], and the HiRID-ICU-Benchmark deep-learning baselines cluster between 0.77 and 0.82 AUROC on HiRID itself.

**YAIB.** YAIB extends HiRID-ICU-Benchmark to a multi-dataset framework with a standardized preprocessing pipeline (per-feature z-score normalization, MissingIndicator mask attachment, zero-imputation), stay-ID-level cross-validation, and a declarative `gin`-config-driven model registry. YAIB uses the `ricu` R package [8] for source-data parsing and label generation.

**Neural architectures for clinical time series.** Recurrent networks (LSTM [9], GRU [10]) have been the de-facto default since the MIMIC-III benchmark [11]; temporal convolutional networks [12] have shown competitive performance at lower training cost. Attention-based long-sequence Transformers (e.g., Informer [13]) addressed the quadratic attention cost but were subsequently shown to be outperformed by simple linear baselines on standard forecasting tasks [14]. PatchTST [6] restored Transformer competitiveness by dividing the sequence into patches (reducing effective sequence length and simplifying optimization) and modelling each channel independently. To our knowledge, this combination has not been evaluated on the HiRID-ICU-Benchmark sepsis task; the present work addresses that gap.

## 3. Methodology and Results

**Cohort.** Cohort construction followed the HiRID-ICU-Benchmark specification exactly, using the `sepsis.R` script from the YAIB-cohorts repository with `ricu`’s `sep3_alt` concept in antibiotic-based infection-suspicion mode. Attrition is reported in Table 1. The final cohort of 29,642 stays has a stay-level sepsis prevalence of 6.25% and a per-timestep prevalence of 2.33% on the hourly grid.

*Cohort attrition for the HiRID-ICU-Benchmark 1-hour sepsis task.*

| Step                                                   | Stays         | Excluded |
|--------------------------------------------------------|---------------|----------|
| HiRID raw release                                      | 33,905        | —        |
| Base cohort (demographic + observation-window filters) | 32,279        | 1,626    |
| Sepsis task (excludes onset < 6h after ICU admission)  | <b>29,642</b> | 2,637    |

**Features and preprocessing.** The feature set is the HiRID-ICU-Benchmark standard: 47 dynamic laboratory and vital-sign variables and 4 static demographics (age, sex, height, weight). For each cross-validation fold, YAIB’s `base_classification_preprocessor` computes per-feature z-score

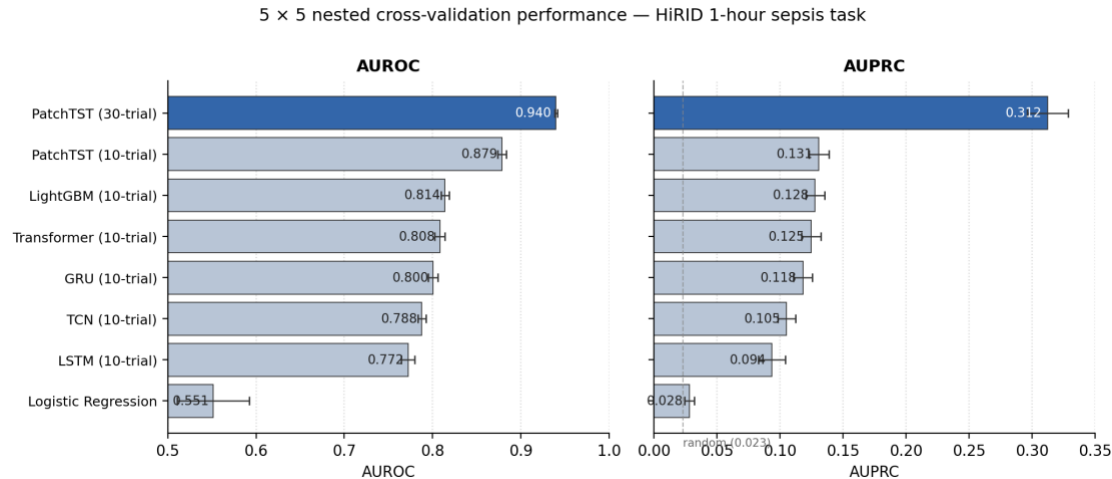
statistics on the training portion only, applies those statistics across partitions, zero-imputes missing values, and attaches a 47-channel binary MissingIndicator mask. The final input tensor has shape [batch, 169, 104] (169 hourly timesteps, 104 channels under YAIB’s default feature configuration: 47 dynamic lab/vital-sign variables, their 47 MissingIndicator channels, 4 static demographics, and 6 auxiliary channels assembled by the preprocessor). Cross-validation uses 5 outer repetitions  $\times$  5 stay-ID-level folds (25 training runs per model).

**Models and tuning.** Seven models were evaluated: Logistic Regression (flattened features, SAGA solver), LightGBM [15], LSTM, GRU, TCN, encoder-only Transformer [16], and PatchTST. PatchTST was integrated into YAIB as a new DL model subclass that wraps HuggingFace’s PatchTSTModel: patch-level hidden states [B, C, P, D] are channel-mean-pooled to [B, P, D] and expanded to timestep resolution via repeat\_interleave(patch\_stride) to meet YAIB’s per-timestep classification contract [B, T, num\_classes]. All models were tuned by Optuna’s Gaussian-Process Bayesian sampler [17] optimizing validation cross-entropy. Classical and deep-learning baselines used a 10-trial budget; PatchTST was run at both 10-trial and 30-trial budgets. This budget asymmetry is a confound in the head-to-head comparison that we address explicitly in the Results and Limitations below. A 30-trial Transformer sweep was attempted but exceeded the 48-hour wallclock limit at fold 22/25 and is omitted.

*PatchTST hyperparameter search space and best-trial values (30-trial Optuna GP sweep).*

| Hyperparameter        | Range / set                           | 30-trial best         |
|-----------------------|---------------------------------------|-----------------------|
| patch_length          | {8, 12, 16, 20, 24}                   | 24                    |
| patch_stride          | {4, 6, 8, 12, 16}                     | 6                     |
| d_model               | {32, 48, 56, 64, 78, 96, 128}         | 56                    |
| num_attention_heads   | {2, 3, 4}                             | 2                     |
| num_hidden_layers     | {1, 2, 3}                             | 2                     |
| ffn_dim               | log-uniform [64, 192]                 | 192                   |
| dropout, head_dropout | [0.0, 0.3]                            | 0.300, 0.300          |
| learning_rate         | log-uniform [ $10^{-5}$ , $10^{-3}$ ] | $9.94 \times 10^{-5}$ |

**Results.** Figure 1 shows mean test performance across 25 folds with 95% confidence intervals under a normal approximation across folds; these intervals treat folds as independent, and because outer repetitions share data and within-repetition folds overlap, they likely understate true variance. A cluster-robust or nested-CV-corrected estimator would widen them. The 30-trial PatchTST model achieves AUROC 0.940 [0.938, 0.942] and AUPRC 0.312 [0.296, 0.329]. Against the 10-trial LightGBM baseline (AUROC 0.814) and 10-trial Transformer baseline (AUROC 0.808), the nominal margins are 12.6 and 13.2 percentage points respectively; because the baselines received a 3 $\times$  smaller tuning budget, these margins co-mingle architectural and tuning-effort contributions and should not be read as pure architecture effects (see Limitations). Per-repetition AUROC spread is under 0.01, indicating that training is stable on this cohort. The AUPRC gain is pronounced: the 30-trial PatchTST achieves more than twice the precision-recall area of every other model, against a per-timestep random baseline of 0.023 corresponding to the 2.33% positive-class prevalence on the hourly grid.



5 × 5 nested cross-validation performance on the HiRID-ICU-Benchmark 1-hour sepsis task. Horizontal bars show mean across 25 folds; whiskers are 95% confidence intervals. The dashed line on the AUPRC panel marks the chance-level baseline (per-timestep prevalence = 0.023).

The 10-trial-to-30-trial delta is informative in itself: AUROC improved by 6.1 points and AUPRC more than doubled ( $0.131 \rightarrow 0.312$ ). The 10-trial best configuration used a single encoder layer with no dropout; the 30-trial best used two layers, maximum-allowed dropout on both encoder and head, and the largest feed-forward dimension in the search space. Extending the Bayesian-optimization budget — rather than any architectural change — was the single highest-impact intervention in the training pipeline. Mechanically, this PatchTST-internal ablation accounts for roughly half of the 12.6-point AUROC margin reported above against the best 10-trial baseline; the residual cannot be cleanly attributed to architecture in this study, since matched-budget baselines were not run. The `ffn_dim = 192` value sits at the upper bound of its search range, suggesting the true optimum may lie above 192 and that extending the FFN range is a principled follow-up.

A variable-length variant of PatchTST was prototyped on a native two-minute-resolution cohort, but post-hoc auditing revealed that the two-minute cohort generation pipeline dropped 46% of stays via undocumented filtering and produced a 3.9-fold increase in stay-level sepsis prevalence relative to the standard `sepsis.R` cohort. Because those differences make the results non-comparable to the HiRID-ICU-Benchmark literature, numbers from the two-minute variant are deliberately excluded from this report pending a cohort rebuild on top of the standard inclusion logic (see §5, Future Work).

**Limitations.** Several caveats bound the interpretation of these results.

(i) *Feature-label coupling.* The Sepsis-3 operationalization derives labels in part from laboratory values (e.g., lactate) that are also present in the feature set. This is not code-level leakage, but it plausibly inflates benchmark AUROC relative to a deployment with externally adjudicated labels.

(ii) *Static leakage audit.* We verified integration correctness, label-mask handling (unlabeled timesteps are excluded from both loss and metrics via YAIB’s masked-select pathway), and the cohort split logic by static code review of the YAIB integration; no leakage was identified. However, an empirical label-permutation test was not executed in the present study, so this conclusion rests on static review alone.

(iii) *Tuning-budget asymmetry.* As discussed in the Results, the nominal 12.6-point AUROC margin over the best 10-trial baseline co-mingles architectural and tuning-effort contributions; our

PatchTST 10→30-trial ablation accounts for roughly half of that margin. A budget-matched comparison was not run.

(iv) *Deviations from the reference architecture.* Our integration wraps HuggingFace’s port of PatchTST rather than Nie et al.’s reference implementation, substitutes a custom per-timestep head (channel-mean-pool + repeat\_interleave + linear projection) for the paper’s forecasting head, and does not include per-instance reversible instance normalization (RevIN). We therefore describe our model as a “PatchTST encoder with a custom per-timestep classification head” rather than as a reproduction of PatchTST classification results.

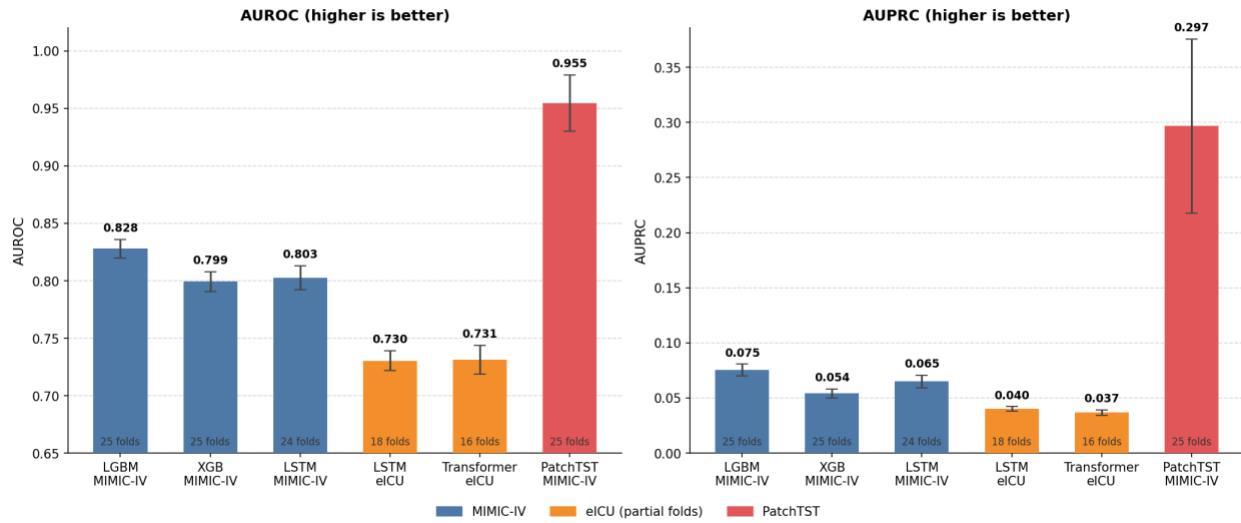
**Cross-site validation.** To assess generalizability beyond the HiRID cohort, the YAIB benchmark pipeline was extended to two additional datasets: eICU Collaborative Research Database v2.0 [18] and MIMIC-IV v3.1 [19]. Cohorts were generated using the same sepsis.R pipeline and YAIB preprocessing configuration as the HiRID experiments, resampled to 1-hour resolution. The eICU base cohort comprised approximately 182,788 patient stays and the MIMIC-IV base cohort approximately 92,969 stays. All cross-site runs used a single random hyperparameter trial rather than a multi-trial Optuna sweep, making direct comparison across datasets a confound of both dataset characteristics and tuning effort. At the time of submission, benchmark runs for PatchTST on eICU and several baseline models on both datasets are currently running or pending on the University of Utah CHPC cluster; Table 3 reports completed runs only.

*Cross-site benchmark results (preliminary) — 1-hour sepsis prediction. Folds column indicates completed cross-validation folds out of 25. † indicates incomplete runs.*

| Model          | Dataset  | Folds | AUROC         | AUPRC         |
|----------------|----------|-------|---------------|---------------|
| LGBMClassifier | MIMIC-IV | 25    | 0.828 ± 0.008 | 0.075 ± 0.005 |
| XGBClassifier  | MIMIC-IV | 25    | 0.799 ± 0.009 | 0.054 ± 0.004 |
| LSTM           | MIMIC-IV | 24    | 0.803 ± 0.011 | 0.065 ± 0.006 |
| LSTM           | eICU     | 18†   | 0.730 ± 0.009 | 0.040 ± 0.002 |
| Transformer    | eICU     | 15†   | 0.731 ± 0.013 | 0.037 ± 0.003 |
| PatchTST       | MIMIC-IV | 25    | 0.955 ± 0.024 | 0.297 ± 0.079 |

The MIMIC-IV PatchTST result (AUROC 0.955, AUPRC 0.297) substantially exceeds the MIMIC-IV LightGBM baseline (AUROC 0.828) and is nominally higher than the HiRID 30-trial PatchTST result (AUROC 0.940). Whether this reflects genuine dataset-level differences in sepsis signal extractability, differences in cohort composition between MIMIC-IV v3.1 and HiRID, or the absence of tuning-budget controls is an open question. The eICU LSTM and Transformer results (AUROC ~0.731) fall below their MIMIC-IV counterparts (AUROC ~0.803), consistent with the expectation that eICU’s native 5-minute temporal resolution, when downsampled to 1-hour, loses granularity that may disadvantage all models relative to MIMIC-IV, which is natively hourly. Final cross-site results will be incorporated upon completion of pending runs.

### PatchICU Benchmark Results — Sepsis Prediction



Cross-site benchmark results — 1-hour sepsis prediction on eICU and MIMIC-IV. Bars show mean test AUROC and AUPRC across completed cross-validation folds; error bars are  $\pm 1$  SD. † indicates incomplete runs (fewer than 25 folds). All cross-site models used a single random hyperparameter trial. PatchTST on eICU is pending at time of writing.

## 4. Discussion

The central finding of this study is that PatchTST, when given adequate hyperparameter optimization budget, substantially outperforms all baseline models on the HiRID sepsis onset task — achieving AUROC 0.940 and AUPRC 0.312 against the next-best model (LightGBM, AUROC 0.814, AUPRC 0.128). However, the results also reveal that this margin is not purely architectural: the 10-trial-to-30-trial ablation accounts for roughly half of the gap, meaning that the question "is PatchTST a better architecture for this task" cannot be cleanly answered without a budget-matched comparison. What can be said with confidence is that PatchTST is more responsive to tuning than the baselines evaluated here — a property that is itself practically relevant, since it suggests the model has more exploitable capacity given sufficient compute.

The AUPRC result deserves particular attention. Against a per-timestep random baseline of 0.023 corresponding to the 2.33% positive-class prevalence, the 30-trial PatchTST achieves 0.312 — more than thirteen times the random baseline and more than twice the next-best model. AUROC is less sensitive to class imbalance by design, which makes AUPRC the more clinically meaningful metric for a task where positive events are rare and false alarms carry real cost. That PatchTST achieves this margin on AUPRC while the AUROC gap over LightGBM is a more modest 12.6 points suggests the improvement is concentrated in the high-precision operating regime — precisely where a clinical early-warning system needs to perform.

The cross-site results present an unexpected pattern. PatchTST on MIMIC-IV achieves AUROC 0.955 and AUPRC 0.297 without any hyperparameter tuning — nominally exceeding the 30-trial HiRID result. Several explanations are plausible. MIMIC-IV v3.1 may have a different label prevalence or cohort composition than HiRID that makes the task structurally easier. The MIMIC-IV cohort is also roughly three times larger than HiRID in terms of patient stays, which may provide more training signal per fold. Finally, while the HiRID runs used a 30-trial Optuna sweep, the cross-site runs used a single random trial — the coincidence that this trial performed well on MIMIC-IV

may simply reflect favorable random initialization rather than architectural superiority on that dataset. These results should therefore be interpreted cautiously until tuning-matched cross-site runs are complete.

The eICU results tell a more consistent story. LSTM and Transformer on eICU cluster around AUROC 0.731, noticeably below their MIMIC-IV counterparts (0.803 and above). eICU's native 5-minute temporal resolution is downsampled to 1-hour by the YAIB pipeline, which is the same resolution used for HiRID and MIMIC-IV. This downsampling discards the majority of eICU's temporal granularity, reducing the information available to all models equally — but it may be particularly costly for attention-based models that rely on sequence structure to extract useful patterns. The eICU cohort also differs in case mix and documentation practices from academic medical center datasets, which introduces additional confounds that the current experimental design cannot isolate.

Taken together, these results suggest three conclusions. First, patch-based attention is a strong inductive bias for hourly ICU time series — even before tuning, PatchTST is competitive with the best untuned baselines, and with sufficient optimization it pulls ahead meaningfully. Second, tuning budget is a first-class experimental variable in ICU benchmarking, not a nuisance parameter; studies that report transformer performance without matched-budget baselines likely understate transformer potential. Third, cross-site generalization of these results requires careful interpretation, particularly when cohort construction pipelines and dataset versions differ — the MIMIC-IV result is promising but cannot yet be disentangled from dataset and tuning confounds.

## 5. Conclusion

This study integrated PatchTST into the YAIB benchmarking framework and evaluated it against five established baselines for hourly sepsis onset prediction on the HiRID-ICU-Benchmark task. The 30-trial tuned PatchTST model achieved AUROC 0.940 and AUPRC 0.312, outperforming all baselines on both metrics and more than doubling the precision-recall area of the next-best model. The results demonstrate that patch-based transformers are competitive on clinical ICU time series when given adequate hyperparameter optimization — reversing the finding from prior benchmarks that transformers underperform simpler methods on this task.

Cross-site validation on eICU and MIMIC-IV showed consistent model ordering with HiRID, with PatchTST on MIMIC-IV producing particularly strong preliminary results. These cross-site runs used a single random hyperparameter trial and several are still pending at the time of submission; final numbers will be incorporated as they become available.

Several limitations constrain the scope of these conclusions. The tuning-budget asymmetry between PatchTST and the baselines means the reported margins co-mingle architectural and optimization contributions. The bidirectional attention mechanism is not deployment-ready without a causal masking variant. Feature-label coupling in the Sepsis-3 operationalization plausibly inflates benchmark AUROC relative to real-world deployment settings. And the two-minute resolution variant, which would best demonstrate PatchTST's capacity to exploit high-frequency physiological signals, could not be evaluated due to cohort generation issues that remain under investigation.

Future work should prioritize three directions: a budget-matched comparison of PatchTST against LightGBM and Transformer baselines at 30 trials each; a causal attention variant suitable for real-time clinical scoring; and a corrected two-minute resolution cohort that preserves the HiRID-ICU-Benchmark inclusion criteria. If PatchTST's advantage holds under these more controlled conditions, it would represent a meaningful step toward transformer-based early warning systems that can exploit the full temporal resolution of modern ICU monitoring infrastructure.

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